

Time-ordered expansion for the derivation of time-convolutionless master equations

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In this work we propose a novel numerical method to compute an approximation of the time-convolutionless master equation by leveraging a series expansion in time. The proposed method is tested against prototypical models of interest including a dephasing spin-boson model a central spin model and an Ising spin chain.

I. INTRODUCTION

For the reader's convenience we next summarize the main contributions of this work:

- In Sec. we propose a novel numerical method to compute an approximate version of the time-convolutionless master equation. Throughout the paper we highlight both the strengths and weaknesses of the proposed approximation procedure. Namely: the proposed procedure is simpler (numerically and analytically) than the alternatives, allows for an initial model that is in Lindblad form (not only purely Hamiltonian) and does not rely on a weak coupling assumption hence it allows to study systems even in strong coupling regimes. On the other hand the proposed reduction procedure relies on an expansion in time and thus provides a good approximation only for small times. While this aspect is limiting, similar restriction also hold in the well known approximations of the time-convolutionless master equation proposed by [1].
- In Sec. IV we prove that, when the original dynamics is purely Hamiltonian, the proposed first and second order approximation provide a CPTP (two-parameter) semigroup.
- In Sec. V we test the proposed procedure on several prototypical examples including: a “classical” linear model, a dephasing spin boson model, a central spin model and and Ising spin chain.

It is important to point out that the method proposed in this work is not intended to substitute the perturbative approach at deriving the time-convolutionless master equation (TCL ME) but rather to provide an alternative that can be of use whenever the main assumptions of the standard method do not hold, e.g. in the strong coupling regime.

II. PROBLEM SETTING AND KNOWN RESULT

A. Problem setting

Let us consider a quantum system defined on a Hilbert space $\mathcal{H}$, and let $\mathfrak{B}(\mathcal{H})$ denote the set of bounded operators acting on $\mathcal{H}$. Unless differently specified, We only

consider finite-dimensional systems, i.e. $\dim(\mathcal{H}) < \infty$, $\mathcal{H} \simeq \mathbb{C}^n$ and $\mathfrak{B}(\mathcal{H}) \simeq \mathbb{C}^{n \times n}$. The state of such a system is described by a density operator $\rho \in \mathfrak{D}(\mathcal{H})$ with $\mathfrak{D}(\mathcal{H}) = \{\rho \in \mathfrak{B}(\mathcal{H}) | \rho = \rho^\dagger \geq 0, \text{tr} \rho = 1\}$. Here, We assume that the state of the system evolves according to the Von-Neumann master equation

$$\dot{\rho}_t = \mathcal{L}(\rho_t) \quad (1)$$

where $\rho_0 \in \mathfrak{D}(\mathcal{H})$ is the system's initial condition, and $\mathcal{L}$ is the celebrated Lindblad (or GKLS) generator [2, 3]:

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_k \mathcal{D}_{L_k}(\rho) \quad (2)$$

where $H = H^\dagger$ is the system's Hamiltonian and $\mathcal{D}_L(\rho) = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$ represents the system's dissipation toward a Markovian environment represented by a set of operators $\{L_k\}$ called noise operators. The evolution of the quantum system is then described by the exponential map $\rho_t = e^{\mathcal{L}t}(\rho_0)$. Importantly, $e^{\mathcal{L}t}$ generates a quantum dynamical semigroup, i.e. a one-parameter semigroup of completely positive (CP) and trace preserving maps (TP). In the following we will say that a super-operator is of *Lindblad-type* if it can be put in the form of Eq. (2).

Note that, while physically the most natural setting is the Hamiltonian one, i.e. $\mathcal{L}(\cdot) = -i[H, \cdot]$, mathematically there is no need to restrict to this case. For this reason, in this work, we start with the general assumption that $\mathcal{L}$ is a generator of Lindblad form and then focus on the Hamiltonian case when necessary.

Next, assume to be given a CPTP, not necessarily orthogonal (w.r.t. $\langle \cdot, \cdot \rangle_{HS}$), projector $\mathcal{P} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathfrak{B}(\mathcal{H})$, $\mathcal{P}^2 = \mathcal{P}$, whose image will be denoted by $\mathcal{V} \equiv \text{Im} \mathcal{P}$. This provides a generalization to the superoperator projector setting considered by [4, 5]. Furthermore, for the sake of simplicity, I assume that $\mathcal{P}$ admits a full rank fixed point, i.e. $\exists \rho \in \mathfrak{D}(\mathcal{H})$, $\rho > 0$ such that $\mathcal{P}(\rho) = \rho$. As we shall see in more details later (in Sec. IV A) such a projector is in fact a conditional expectation and enjoys some useful properties. For the moment however, it is sufficient to know that $\mathcal{P}$ can be decomposed in two factors $\mathcal{P} = \mathcal{J}\mathcal{R}$ which take the name of reduction and injection maps $\mathcal{R} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathfrak{B}(\mathcal{K})$ and $\mathcal{J} : \mathfrak{B}(\mathcal{K}) \rightarrow \mathfrak{B}(\mathcal{H})$ such that $\mathcal{R}\mathcal{J} = \mathcal{I}_{\mathcal{K}}$ and $\text{Im} \mathcal{J} = \mathcal{V}$ $\text{Im} \mathcal{R} = \mathcal{A}$ and where $\mathcal{A} \subseteq \mathfrak{B}(\mathcal{K})$ is a $*$ -subalgebra of $\mathfrak{B}(\mathcal{K})$ for some Hilbert space $\mathcal{K}$, $\dim(\mathcal{K}) \leq \dim(\mathcal{H})$. For later convenience we shall also define the projector $\mathcal{Q} \equiv \mathcal{I} - \mathcal{P}$. Note that neither $\mathcal{P}$ or $\mathcal{Q}$ need to be orthogonal, i.e. self adjoint.

To provide some intuition on these mathematical objects we anticipate the reduction and injection maps we are going to use in the applications of Sec. V:

- Assume a bipartite case, i.e. $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. We can then consider $\mathcal{P}(\rho) = \text{tr}_B(\rho) \otimes \tau$ for some $\tau \in \mathfrak{D}(\mathcal{H}_B)$, $\tau > 0$ to be our CPTP projector. Then, by choosing $\mathcal{K} = \mathcal{H}_S$ and $\mathcal{A} = \mathfrak{B}(\mathcal{H}_S)$ we have the maps $\mathcal{R}(\rho) = \text{tr}_B(\rho)$ and $\mathcal{J}(\mu) = \mu \otimes \tau$.
- Another CPTP projector we might want to consider is the projector onto the diagonal matrix $\mathcal{P}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ in which case $\mathcal{K} = \mathcal{H}$, $\mathcal{A}$ is the commutative algebra given by $\mathcal{A} \equiv \text{span}\{|j\rangle\langle j|, j = 0, \dots, n-1\} \simeq \mathbb{C}^n$, and $\mathcal{R}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ in which case $\mu = \mathcal{R}(\rho) \in \mathbb{C}^n$, and $\mathcal{J}(\mu) = \sum_{j=0}^{n-1} |j\rangle\langle j| \mu \langle j|$ factorize $\mathcal{P}$.

The reduction map $\mathcal{R}$ determines the reduced state – the one we want to keep track of: Let $\eta_t \in \mathfrak{D}(\mathcal{K})$ be defined as $\eta_t = \mathcal{R}(\rho_t)$ for all $t \geq 0$. The evolution of the reduced state is then given by

$$\eta_t = \mathcal{R}e^{\mathcal{L}t}(\rho_0) = \mathcal{R}e^{\mathcal{L}t}\mathcal{J}(\eta_0) + \mathcal{R}e^{\mathcal{L}t}\mathcal{Q}(\rho_0).$$

We next introduce an important simplifying assumption that will hold throughout the rest of the paper and that it is often assumed in similar situations, see e.g. [6]

Assumption 1. We assume that $\mathcal{P}(\rho_0) = \rho_0$.

This assumption directly implies that the evolution of the reduced state takes the form:

$$\eta_t = \mathcal{R}e^{\mathcal{L}t}\mathcal{J}(\eta_0). \quad (3)$$

In this work we are interested in characterizing a time-dependent generator that approximate well the evolution of η_t for $t \geq 0$. To that aim, let us define a density operator $\mu_t \in \mathfrak{D}(\mathcal{H})$ whose evolution is described by

$$\begin{cases} \dot{\mu}_t = \mathcal{F}_t(\mu_t) \\ \mu_0 = \eta_0 \end{cases} \quad (4)$$

and where $\mathcal{F}_t$ is a time-dependent generator. Our aim is then to construct $\mathcal{F}_t$ so that $\mu_t \approx \eta_t$. In other words we want to determine $\mathcal{F}_t$ so that

$$\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} \approx \mathcal{R}e^{\mathcal{L}t}\mathcal{J}$$

where $\mathbb{T}$ denotes the time-ordering operator. The sense in which the approximation $\approx$ is intended shall be defined later.

Desirable properties for $\mathcal{F}_t$ certainly include the fact that we would like $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$ to be CPTP for all $t \geq 0$. Note that necessary conditions to impose on $\mathcal{F}_t$ so that $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$ is CPTP for all $t \geq 0$ are not known [7].

A graphical representation of the approximation problem is given in Fig. 1.

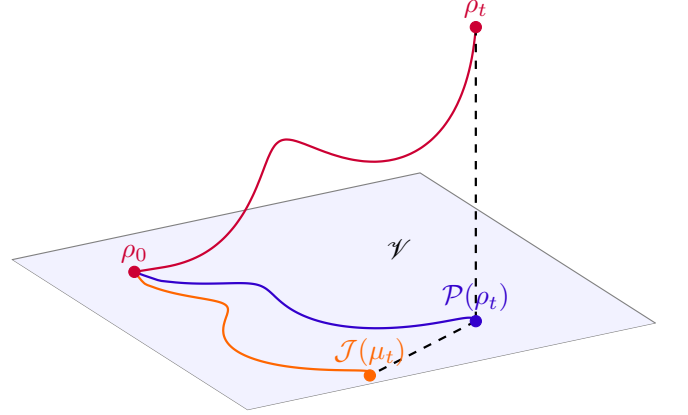


Figure 1: Graphical representation of the approximation problem. The pale blue plane represents the operator space $\mathcal{V}$ onto which $\mathcal{P}$ projects. The red curve represents the trajectory of the original model ρ_t , which starts in $\mathcal{V}$ at $t = 0$. The blue curve represents the projection of the true trajectory $\mathcal{P}(\rho_t)$, i.e. the one we want to approximate. Finally, the orange curve represents the approximate evolution we aim to find, i.e. $\mathcal{J}(\mu_t)$.

B. Time-convolutionless master equation

Before moving to the method proposed in this work for computing the time dependent generator $\mathcal{F}_t$ we shall discuss the derivation of time-convolutionless master equations, which is the standard method found in the literature [1, 8–10] to solve the issue at hand.

Dividing the evolution of ρ_t in the evolution of $\mathcal{P}\rho_t$ and that of $\mathcal{Q}\rho_t$ we have

$$\frac{d}{dt}\mathcal{P}\rho_t = \mathcal{P}\mathcal{L}\mathcal{P}\rho_t + \mathcal{P}\mathcal{L}\mathcal{Q}\rho_t, \quad (5)$$

$$\frac{d}{dt}\mathcal{Q}\rho_t = \mathcal{Q}\mathcal{L}\mathcal{P}\rho_t + \mathcal{Q}\mathcal{L}\mathcal{Q}\rho_t. \quad (6)$$

Solving the second equation for $\mathcal{Q}\rho_t$ we obtain

$$\mathcal{Q}\rho_t = e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + \int_0^t e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s}\mathcal{Q}\mathcal{L}\mathcal{P}\rho_{t-s}ds$$

and substituting this equation into the first equation we obtain the famous Nakajima-Zwanzig equation [11, 12]:

$$\begin{aligned} \frac{d}{dt}\mathcal{P}\rho_t &= \mathcal{P}\mathcal{L}\mathcal{P}\rho_t + \mathcal{P}\mathcal{L}e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 \\ &\quad + \int_0^t \mathcal{P}\mathcal{L}\mathcal{Q}e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s}\mathcal{Q}\mathcal{L}\mathcal{P}\rho_{t-s}ds. \end{aligned} \quad (7)$$

In order to derive the time-convolutionless master equation we recall that $\rho_{t-s} = e^{-\mathcal{L}s}\rho_t$ so that

$$\mathcal{Q}\rho_t = e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + \int_0^t e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s}\mathcal{Q}\mathcal{L}\mathcal{P}e^{-\mathcal{L}s}ds \cdot (\mathcal{P} + \mathcal{Q})\rho_t.$$

Now, defining

$$\mathcal{M}_t \equiv \mathcal{I} - \int_0^t e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s} \mathcal{Q}\mathcal{L}\mathcal{P} e^{-\mathcal{L}s} ds$$

we have

$$\mathcal{M}_t \mathcal{Q}\rho_t = e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t} \mathcal{Q}\rho_0 + (\mathcal{I} - \mathcal{M}_t) \mathcal{P}\rho_t.$$

If we then assume that $\mathcal{M}_t$ is invertible we obtain

$$\mathcal{Q}\rho_t = \mathcal{M}_t^{-1} e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t} \mathcal{Q}\rho_0 + (\mathcal{M}_t^{-1} - \mathcal{I}) \mathcal{P}\rho_t.$$

Substituting this in Eq. (5) we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{P}\rho_t &= \mathcal{P}\mathcal{L}\mathcal{P}\rho_t + \mathcal{P}\mathcal{L}\mathcal{M}_t^{-1} e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t} \mathcal{Q}\rho_0 \\ &\quad + \mathcal{P}\mathcal{L}(\mathcal{M}_t^{-1} - \mathcal{I}) \mathcal{P}\rho_t \\ &= \mathcal{P}\mathcal{L}\mathcal{M}_t^{-1} e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t} \mathcal{Q}\rho_0 + \mathcal{P}\mathcal{L}\mathcal{M}_t^{-1} \mathcal{P}\rho_t. \end{aligned} \quad (8)$$

Note that under the assumptions that $\mathcal{P}\rho_0 = \rho_0$ we have that $\mathcal{Q}\rho_0 = 0$ and thus Eq. (8) gets simplified to

$$\frac{d}{dt} \mathcal{P}\rho_t = \underbrace{\mathcal{P}\mathcal{L}\mathcal{M}_t^{-1} \mathcal{P}}_{\equiv \mathcal{K}_t} \rho_t \quad (9)$$

which, separating the Projector $\mathcal{P}$ into its factors $\mathcal{R} = \mathcal{J}\mathcal{R}$, gets simplified to

$$\frac{d}{dt} \eta_t = \underbrace{\mathcal{R}\mathcal{L}\mathcal{M}_t^{-1} \mathcal{J}}_{\equiv \mathcal{F}_t} \eta_t. \quad (10)$$

A few comments on this solution are in order: i) The invertibility of $\mathcal{M}_t$ needs to be verified and typically only holds for small t , see e.g. [6]. ii) Note that, in order to compute $\mathcal{F}_t$ (or equivalently $\mathcal{K}_t$) one needs to first compute the integral that defines $\mathcal{M}_t$ and then to compute the inverse of $\mathcal{M}_t$. Both these tasks are challenging (numerically and analytically) and it is generally considered that solving this integral and inverse is as difficult as solving the original model (1), see e.g. [5]. Nevertheless, Eq. (8) provides an excellent starting point to compute an approximation of the reduced evolution.

To circumvent these numerical issues, $\mathcal{K}_t$ is typically approximated by expanding it as an expansion in the coupling strength ε and by computing the terms of the expansion as ordered cumulants [4, 6, 13]. In the typical setting, the system is considered to be bipartite, i.e. $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$ and the dynamics purely Hamiltonian, i.e. $\mathcal{L}(\rho) = -i[H, \rho]$ with Hamiltonian $H = H_0 + \varepsilon V$ where H_0 is the free Hamiltonian of the system and bath, V is the interaction Hamiltonian and ε is the coupling strength. Under these assumptions, The time dependent generator $\mathcal{K}_t$ is expressed as

$$\mathcal{K}_t = \sum_{n=1}^{\infty} \mathcal{K}_n(t)$$

where

$$\begin{aligned} \mathcal{K}_n(t) &= \int_0^t dt_1 \int_0^{t_1} dt_2 \cdots \int_0^{t_{n-2}} dt_{n-1} \\ &\quad \times \sum_q (-1)^q \mathcal{P}\mathcal{L}_t \cdots \mathcal{L}_{t_i} \mathcal{P}\mathcal{L}_{t_j} \cdots \mathcal{L}_{t_k} \mathcal{P} \cdots \mathcal{P} \end{aligned}$$

where $\mathcal{L}_t = -i[H(t), t]$ and $H(t)$ is the Hamiltonian in rotating frame with respect to H_0 . The series is then approximated by stopping at a given order N , i.e. $\mathcal{K}_t \approx \sum_{n=1}^N \mathcal{K}_n(t)$.

In many cases of interest this approximation of the TCL-ME provides a good approximation of the open quantum system dynamics, both in the long-time and short-time regime. It is however important to note that there is no guarantee that such an approximation holds also for long times. As an example of this, see [5] where it is shown that TCL2 and TCL4 (approximations with $N = 2, 4$) do not approximate the true dynamics in the long time regime.

III. POLYNOMIAL APPROXIMATIONS OF THE TIME-DEPENDENT GENERATOR

This work leverages a key assumption: we assume that the time dependent generator $\mathcal{F}_t$ is polynomial in t , i.e. $\mathcal{F}_{t,N} \equiv \sum_{k=0}^N t^k \mathcal{F}_{(k+1)}$. This assumption is clearly restrictive, but it greatly simplifies the computation of the time-ordered exponential $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds}$ as shown in the next key result.

Theorem 1. *Let us assume that $\mathcal{F}_{t,N} = \sum_{k=0}^N t^k \mathcal{F}_{(k+1)}$. Then the time-ordered exponential can be written as*

$$\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} = \sum_{k=0}^{+\infty} t^k \mathcal{E}_{(k)} \quad (11)$$

where $\mathcal{E}_{(0)} = \mathcal{I}$ and

$$\mathcal{E}_{(k)} = \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} \quad (12)$$

and where we used the convention $\mathcal{F}_{(k)} = 0$ for all $k > N$.

Proof of this result and an explanatory example are given in Appendix A 1.

Note that Eq. (12) provides a closed formula for the coefficients $\mathcal{E}_{(k)}$ of the expansion of $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds}$ as a function of the coefficients $\mathcal{F}_{(k)}$. Furthermore, only the terms $\{\mathcal{E}_{(\ell)}\}_{\ell=1}^{k-1}$ appear in Eq. (12), which means that this equation can be used iteratively to compute each term $\mathcal{E}_{(k)}$.

Theorem 1 provides the main intuition on how to construct the polynomial time dependent generator $\mathcal{F}_{t,N}$: Eq. (11) provides an expansion for $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds}$ in the time

variable t . We can then expand $e^{t\mathcal{L}}$ in t and construct $\mathcal{F}_{(k)}$ so that the terms of $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds}$ match the terms of $\mathcal{R}e^{t\mathcal{L}}\mathcal{J}$ by orders of t , similarly to the procedure used in [14]. The resulting procedure, which represents the main contribution of this work, is stated next.

Theorem 2. *Let us define*

$$\mathcal{F}_{t,N} \equiv \sum_{k=0}^N t^k \mathcal{F}_{(k+1)} \quad (13)$$

where

$$\mathcal{F}_{(k)} \equiv k \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} - \sum_{h=1}^{k-1} \mathcal{F}_{(k-h)} \frac{\mathcal{R}\mathcal{L}^h \mathcal{J}}{h!}. \quad (14)$$

Then:

$$\mathcal{R}e^{\mathcal{L}t} \mathcal{J} - \mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} = O(t^{N+1}). \quad (15)$$

Before proceeding to prove this result we shall observe a few key aspects. First, Eq. (15) finally gives sense to the approximation symbol $\approx$ used in Sec. II i.e. an approximation of the type $O(t^{N+1})$. This implies that the approximation holds only for small times. Second, it is important to notice that the right-hand side of equation (14) depends on $\mathcal{R}\mathcal{L}^k \mathcal{J}$ and $\{\mathcal{F}_{(s)}\}_{s=1}^{k-1}$. This means that equation (14) can be used iteratively to construct the terms $\mathcal{F}_{(k)}$ starting from $k=1$ and growing up to $k=N$. For the reader's convenience the first few terms of the series are reported next:

$$\begin{aligned} \mathcal{F}_{(1)} &= \mathcal{R}\mathcal{L}\mathcal{J} \\ \mathcal{F}_{(2)} &= \mathcal{R}\mathcal{L}^2 \mathcal{J} - \mathcal{F}_{(1)} \mathcal{R}\mathcal{L}\mathcal{J} = \mathcal{R}\mathcal{L}^2 \mathcal{J} - (\mathcal{R}\mathcal{L}\mathcal{J})^2 \\ \mathcal{F}_{(3)} &= \frac{\mathcal{R}\mathcal{L}^3 \mathcal{J}}{2} - \mathcal{F}_{(2)} \mathcal{R}\mathcal{L}\mathcal{J} - \mathcal{F}_{(1)} \frac{\mathcal{R}\mathcal{L}^2 \mathcal{J}}{2} \\ &= \frac{\mathcal{R}\mathcal{L}^3 \mathcal{J}}{2} - \frac{\mathcal{F}_{(1)} \mathcal{F}_{(2)}}{2} - \frac{\mathcal{F}_{(1)}^3}{2} - \mathcal{F}_{(2)} \mathcal{F}_{(1)} \end{aligned}$$

It is interesting to notice a couple of points about the first few terms: i) As one would expect, the term of order one $\mathcal{F}_{(1)}$ coincides with the reduction of the original dynamics which is often used as rough time-independent approximation of the reduced evolution [CITE]; ii) The second order term does not only contain the reduction of the generator squared $\mathcal{R}\mathcal{L}^2 \mathcal{J}$ but also contains a correction term depends on the first order term $\mathcal{F}_{(1)}$.

We are finally ready to prove Theorem 2.

Proof (Proof of Theorem 2). First of all we can use Theorem 1 to write

$$\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} = \sum_{f=0}^{+\infty} t^f \mathcal{E}_{(f)}$$

where $\mathcal{E}_{(k)} = \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} = \frac{1}{k} \sum_{h=0}^{k-1} \mathcal{F}_{(k-h)} \mathcal{E}_{(h)}$

and where we reversed the order of the summation by imposing $h = k - s$. We next prove that, for $1 \leq k \leq N$, we have $\mathcal{E}_{(k)} = \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!}$ by induction. The case $k=1$ is trivial as $\mathcal{E}_{(1)} = \mathcal{F}_{(1)} \mathcal{E}_{(0)} = \mathcal{R}\mathcal{L}\mathcal{J}$. Let then assume that $\mathcal{E}_{(h)} = \frac{\mathcal{R}\mathcal{L}^h \mathcal{J}}{h!}$ for all $h < k$ and let us prove that $\mathcal{E}_{(k)} = \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!}$. Then

$$\begin{aligned} \mathcal{E}_{(k)} &= \frac{1}{k} \sum_{h=0}^{k-1} \mathcal{F}_{(k-h)} \mathcal{E}_{(h)} = \frac{\mathcal{F}_{(k)}}{k} + \frac{1}{k} \sum_{h=1}^{k-1} \mathcal{F}_{(k-h)} \mathcal{E}_{(h)} \\ &= \frac{\mathcal{F}_{(k)}}{k} + \frac{1}{k} \sum_{h=1}^{k-1} \mathcal{F}_{(k-h)} \frac{\mathcal{R}\mathcal{L}^h \mathcal{J}}{h!} = \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} \end{aligned}$$

where we simply substituted the definition of $\mathcal{F}_{(k)}$ given in eq. (14).

By leveraging the definition of the exponential map $e^{\mathcal{L}t}$ we can expand the term $\mathcal{R}e^{\mathcal{L}t} \mathcal{J}$ to obtain

$$\mathcal{R}e^{\mathcal{L}t} \mathcal{J} = \sum_{k=0}^{+\infty} \frac{t^k}{k!} \mathcal{R}\mathcal{L}^k \mathcal{J}.$$

Combining the two terms we obtain

$$\begin{aligned} \mathcal{R}e^{\mathcal{L}t} \mathcal{J} - \mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} &= \sum_{k=0}^{+\infty} t^k \left(\frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} - \mathcal{E}_{(k)} \right) \\ &= \sum_{k=0}^N t^k \left(\frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} - \mathcal{E}_{(k)} \right) \\ &\quad + O(t^{N+1}) \end{aligned}$$

which concludes the proof. $\square$

Remark. Note that in this section we have derived an approximation method that is applicable to any LTI system. In fact, in both Theorem 1 and Theorem 2, no assumption on the structure of $\mathcal{L}$ or $\mathcal{P}$ was used, implying that those results can be used to derive approximate models for “classical” linear systems as well, e.g. in cases where $\rho_t \in \mathbb{C}^n$, $\mathcal{L}, \mathcal{P} \in \mathbb{C}^{n \times n}$ and $\mathcal{F}_t \in \mathbb{C}^{m \times m}$ where $m = \dim \text{Im} \mathcal{P}$. To showcase this point, in Sec. V A we study the problem of reducing a classical linear time-invariant model using the methods presented in this section.

A. Connection with the literature

While Eq. (14) has been derived independently, it shall be noted that a quite similar recursive formula is known in the literature. Specifically [13, Eq. (38)] (but also [15, Eq. (2)] [16, Eq. (2)], [17, Eq. (16)]) provides a recursive formula, similar to (14) for the terms of a Dyson expansion for the time-dependent generator $\mathcal{F}_t$. That formula however was derived starting from an initial

time-dependent model and thus the terms of the recursive formula contain time-ordered integrals that need to be solved in order to compute the terms of the expansion. Because in this work we start from a time-independent model, Eq. (14) does not contain time-ordered integrals, resulting in overall simpler computations.

Another aspect that separates this work from most of the literature, e.g. [8, 18, 19] is the fact that instead of expanding in a power-series of a perturbation parameter ε (which typically represents the coupling strength between system and environment) here we expand in the time variable t . This fact simplifies the calculations and lifts the weak-coupling assumption that is typically introduced but comes at a cost: the approximate model holds only for small times.

While this consequence is clearly limiting, this situation is comparable to the classical derivation of TCL ME as these also hold for small times. **Argument better.**

IV. COMPLETE POSITIVITY OF THE APPROXIMATE MODEL AT FIRST AND SECOND ORDER

In this section we focus on the complete positivity of the reduced dynamics.

A. First order

In this section we recall relevant known results about CPTP projectors, their images and reduced dynamics onto distorted algebras from the literature [20–26]. The consequences of the assumptions we made about the projector $\mathcal{P}$ are summarized in the following proposition.

Proposition 1. *Let $\mathcal{P}$ be a CPTP projector with a full rank fixed point $\bar{\rho} > 0$, $\mathcal{P}(\bar{\rho}) = \bar{\rho}$ and let $\mathcal{V}$ be its image, i.e. $\mathcal{V} \equiv \text{Im}\mathcal{P} \subseteq \mathfrak{B}(\mathcal{H})$. Then there exist a decomposition of the Hilbert space*

$$\mathcal{H} = \bigoplus_k (\mathcal{H}_{F,k} \otimes \mathcal{H}_{G,k}),$$

a unitary operator $U \in \mathfrak{B}(\mathcal{H})$, and a set of density operators $\tau_k \in \mathfrak{D}(\mathcal{H}_{G,k})$ such that

$$\mathcal{V} = U \left(\bigoplus_k \mathfrak{B}(\mathcal{H}_{F,k}) \otimes \tau_k \right) U^\dagger. \quad (16)$$

Furthermore $\mathcal{V}$ is a $$ -algebra closed with respect to the product $X \cdot_\sigma Y \equiv X\sigma Y$ where*

$$\sigma = U \left(\bigoplus_k \mathbb{1}_{F,k} \otimes \tau_k^{-1} \right) U^\dagger.$$

Let then W_k be a linear operator from $\mathcal{H}$ onto $\mathcal{H}_{F,k} \otimes \mathcal{H}_{G,k}$ such that $W_k W_k^\dagger = \mathbb{1}_{F,k} \otimes \mathbb{1}_{G,k}$ [21]. Then the CPTP projection $\mathcal{P}$ onto $\mathcal{V}$ takes the form

$$\mathcal{P}(X) = U \left[\bigoplus_k \text{tr}_{\mathcal{H}_{G,k}} (W_k X W_k^\dagger) \otimes \tau_k \right] U^\dagger. \quad (17)$$

and is orthogonal with respect to the inner product $\langle X, Y \rangle_{\sigma, \lambda} = \text{tr}[X^\dagger \sigma^\lambda Y \sigma^{1-\lambda}]$ for $\lambda \in [0, 1]$.

Proof of this result can be found in [20] and [21, 22, 26–28]. Note that projectors of this type take the name of state extensions and are the dual of conditional expectations [29]. Furthermore, note that a $*$ -algebra closed with respect to a modified product often takes the name of *distorted algebra* [24], while the block structure provided in Eq. (16) takes the name of *Wedderburn decomposition*.

The previous result provides us with a structure for the image of a CPTP projection whenever there exists a full rank state $\bar{\rho}$ such that $\mathcal{P}(\bar{\rho}) = \bar{\rho}$. From this structure one can observe that a distorted algebra $\mathcal{V}$ is isomorphic to a smaller representation which we denote $\mathcal{A}$ where the weighted copies are removed, i.e. $\mathcal{A} \equiv \bigoplus_k \mathfrak{B}(\mathcal{H}_{F,k})$. This allows us to find the two factors $\mathcal{R} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathcal{A}$ and $\mathcal{J} : \mathcal{A} \rightarrow \mathfrak{B}(\mathcal{H})$ such that, $\mathcal{R}\mathcal{J} = \mathcal{I}_{\mathcal{A}}$ and $\mathcal{P} = \mathcal{J}\mathcal{R}$. Furthermore, one can observe [25] that these two factors can be chosen to be CPTP, i.e. for all $X \in \mathfrak{B}(\mathcal{H})$

$$\mathcal{R}(X) = \bigoplus_k \text{tr}_{\mathcal{H}_{G,k}} (W_k X W_k^\dagger) = \bigoplus_k X_{F,k} = \check{X}, \quad (18)$$

and for all $\check{X} \in \mathcal{A}$, $\check{X} = \bigoplus_k X_{F,k}$,

$$\mathcal{J}(\check{X}) = U \left(\bigoplus_k X_{F,k} \otimes \tau_k \right) U^\dagger. \quad (19)$$

The reduction and injection maps given in equations (18) and (19) then allow to compute the reduced dynamics $\mathcal{R}\mathcal{L}\mathcal{J}$ restricted onto the algebra $\mathcal{A}$. One can then prove that the reduced generator $\check{\mathcal{L}}$ is GKLS by leveraging the structure of the reduction and injection maps. We here report [25, Theorem 4] for completeness.

Theorem 3. *Let $\mathcal{P}$ be a CPTP projection with a full rank fixed point, let $\mathcal{V}$ be its image $\mathcal{V} = \text{Im}\mathcal{P} \subseteq \mathfrak{B}(\mathcal{H})$, and let $\mathcal{R}$ and $\mathcal{J}$ denote the CPTP factorization of $\mathcal{P} = \mathcal{J}\mathcal{R}$, as defined in Eq. (18) and (19). Then for any Lindblad generator $\mathcal{L}$, its restriction to $\mathcal{A}$, $\mathcal{R}\mathcal{L}\mathcal{J}$, is also a Lindblad generator, that is, $\mathcal{R}\mathcal{L}\mathcal{J} : \mathcal{A} \rightarrow \mathcal{A}$ and $\{e^{\check{\mathcal{L}}t}\}_{t \geq 0}$ is a quantum dynamical semi-group.*

Note that this theorem only proves the fact that $\mathcal{R}\mathcal{L}\mathcal{J}$ is of Lindblad type. In order to compute the reduced Hamiltonian and noise operators one can use the procedure described in [30, Appendix].

B. Second order: The case of purely Hamiltonian evolution

While the results we derived up to this point hold for general Lindblad generators $\mathcal{L}$, we shall now focus on the special case of purely Hamiltonian evolution, i.e. $\mathcal{L}(\cdot) = -i[H, \cdot]$. Other than the fact that this assumption is interesting in practical cases, we shall see that under this simplifying assumption we are able to prove that the second order term $\mathcal{F}_{(2)}$ is also of Lindblad-type.

The next result is based on the fact that the squared of a purely Hamiltonian generator is equal to a dissipator:

$$\begin{aligned}\mathcal{L}^2(\rho) &= -[H[H, \rho]] = -(H^2\rho - 2H\rho H + \rho H^2) \\ &= 2H\rho H - \{H^2, \rho\} = \mathcal{D}_{\sqrt{2}H}(\rho).\end{aligned}\quad (20)$$

This fact allows us to prove the next result.

Theorem 4. *Under the assumptions of Theorem 3 we have that $\mathcal{F}_{(2)} = \mathcal{R}\mathcal{L}^2\mathcal{J} - (\mathcal{R}\mathcal{L}\mathcal{J})^2$ is of Lindblad-type and for $\mathcal{F}_{t,2} \equiv \mathcal{F}_{(1)} + t\mathcal{F}_{(2)}$ we have that $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,2} ds}$ forms a set of CPTP maps for all $t \geq 0$.*

Proof. By leveraging [25, Proposition 5] we have that $\mathcal{R}\mathcal{L}\mathcal{J}(\check{\rho}) = -i[\check{H}, \check{\rho}]$ where $\check{H} = \mathcal{J}^\dagger(H)$ hence, using equation (20) we obtain $(\mathcal{R}\mathcal{L}\mathcal{J})^2(\check{\rho}) = \mathcal{D}_{\sqrt{2}\check{H}}(\check{\rho})$. On the other hand, by leveraging [30, Proposition 6 and Proposition 7], we have that $\mathcal{R}\mathcal{D}_{\sqrt{2}H}\mathcal{J}(\check{\rho}) = \sum_{\check{C} \in \mathcal{X}(\sqrt{2}H)} \mathcal{D}_{\check{C}}(\check{\rho})$ for a set of operators $\mathcal{X}(\sqrt{2}H) \subset \mathfrak{B}(\mathcal{K})$. Importantly, by [30, Corollary 4], we have that $\mathcal{J}^\dagger(H) \in \mathcal{X}(\sqrt{2}H)$ and thus we can write $\mathcal{R}\mathcal{D}_{\sqrt{2}H}\mathcal{J}(\check{\rho}) = \sum_{\check{C} \in \mathcal{X}(\sqrt{2}H)/\{\sqrt{2}\check{H}\}} \mathcal{D}_{\check{C}}(\check{\rho}) + \mathcal{D}_{\sqrt{2}\check{H}}(\check{\rho})$ since $\sqrt{2}\check{H} = \mathcal{J}^\dagger(\sqrt{2}H)$. Then, by combining these two results we obtain

$$\mathcal{F}_{(2)} = \mathcal{R}\mathcal{L}^2\mathcal{J} - (\mathcal{R}\mathcal{L}\mathcal{J})^2 = \sum_{\check{C} \in \mathcal{X}(\sqrt{2}H)/\{\sqrt{2}\check{H}\}} \mathcal{D}_{\check{C}}(\check{\rho})$$

which is the sum of a set of GKLS, purely dissipative generators and thus of Lindblad type. Combining this with Theorem 3 it trivially follows that $\mathcal{F}_{t,2}$ is of Lindblad type for all $t \geq 0$. $\square$

C. Second order in the special case of bipartite systems

Because the proof of Theorem 4 is based on a few very technical results, we here want to present an equivalent pedagogical derivation of the same result. Note that this derivation is quite relevant in practice as it focuses on the most studied case of a bi-partite system but the cases included in the assumptions of Theorem 4 are more general. This exact case will be the focus of the two applications of interest studied in Sec V B and V C.

Let us assume that $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ and let assume that the CPTP projector $\mathcal{P}$ is given by the $\mathcal{P}(\cdot) = \text{tr}_E(\cdot) \otimes \tau$ for some full rank state $\tau \in \mathfrak{D}(\mathcal{H}_E)$. The projector $\mathcal{P}$ is then factorized into the reduction map $\mathcal{R}(\cdot) = \text{tr}_E(\cdot)$ and the injection map $\mathcal{J}(\cdot) = \cdot \otimes \tau$. In line with the assumptions of this section we assume that $\mathcal{L}(\cdot) = -i\text{ad}_H(\cdot)$. Given the bipartite structure of the Hilbert space $\mathcal{H}$ we can always write $H = \sum_k S_k \otimes E_k$ for two sets of Hermitian operators $\{S_k\} \subset \mathfrak{B}(\mathcal{H}_S)$ and $\{E_k\} \subset \mathfrak{B}(\mathcal{H}_E)$.

We can then proceed to compute the reduced generator approximated at second order $\mathcal{F}_{t,2}(\mu)$. First, using [25, Proposition 5], we have that the generator at first order is

$$\mathcal{F}_{(1)}(\mu) = \mathcal{R}\mathcal{L}\mathcal{J}(\mu) = -i[\check{H}, \mu]$$

where

$$\check{H} = \mathcal{J}^\dagger(H) = \text{tr}_E[(\mathbb{1}_S \otimes \tau)H] = \sum_k a_k(\tau) S_k$$

and $a_k(\tau) \equiv \text{tr}[\tau E_k]$. Next, we can proceed to compute $\mathcal{F}_{(2)}$ by first computing $\mathcal{R}\mathcal{L}^2\mathcal{J}$:

$$\begin{aligned}
\mathcal{RL}^2 \mathcal{J}(\mu) &= \sum_{j,k} \text{tr}_E [2(S_j \otimes E_j)(\mu \otimes \tau)(S_k \otimes E_k) - (S_k \otimes E_k)(S_j \otimes E_j)(\mu \otimes \tau) - (\mu \otimes \tau)(S_k \otimes E_k)(S_j \otimes E_j)] \\
&= \sum_{j,k} 2S_j \mu S_k \underbrace{\text{tr}[E_j \tau E_k]}_{\equiv c_{j,k}(\tau)} - S_k S_j \mu \underbrace{\text{tr}[E_k E_j \tau]}_{=c_{j,k}(\tau)} - \mu S_k S_j \underbrace{\text{tr}[\tau E_k E_j]}_{=c_{j,k}(\tau)} \\
&= \sum_{j,k} c_{j,k}(\tau) [2S_j \mu S_k - \{S_k S_j, \mu\}].
\end{aligned}$$

On the other hand, since $(\mathcal{RL}\mathcal{J})^2 = \mathcal{D}_{\sqrt{2}\tilde{H}}$, we have

$$(\mathcal{RL}\mathcal{J})^2(\mu) = \sum_{j,k} a_j(\tau) a_k(\tau) [2S_j \mu S_k - \{S_k S_j, \mu\}]$$

and thus

$$\begin{aligned}
\mathcal{F}_{(2)}(\mu) &= [\mathcal{RL}^2 \mathcal{J} - (\mathcal{RL}\mathcal{J})^2](\mu) = \\
&= \sum_{j,k} [\chi(\tau)]_{j,k} [2S_j \mu S_k - \{S_k S_j, \mu\}]
\end{aligned}$$

where we defined the matrix $\chi(\tau)$ whose elements are $[\chi(\tau)]_{j,k} \equiv c_{j,k}(\tau) - a_j(\tau) a_k(\tau)$. One can then notice that the matrix $\chi(\tau)$ coincides with a covariance matrix, i.e.

$$\begin{aligned}
[\chi(\tau)]_{j,k} &= \text{tr}[E_j \tau E_k] - \text{tr}[E_j \tau] \text{tr}[E_k \tau] \\
&= \mathbb{E}_\tau[(E_k - \mathbb{E}_\tau[E_k] \mathbb{1}_E)(E_j - \mathbb{E}_\tau[E_j] \mathbb{1}_E)]
\end{aligned}$$

hence $\chi(\tau)$ is a positive semi-definite matrix. Since $\chi(\tau)$ is positive semi-definite, it is diagonalizable with a unitary matrix, i.e. $UU^\dagger = U^\dagger U = \mathbb{1}$ such that $U\chi(\tau)U^\dagger = \gamma(\tau)$ where $\gamma(\tau)$ is a diagonal matrix of positive values (note that in reality U is also a function of τ). We can then define

$$L_h = \sum_j \overline{[U]_{h,j}} S_j$$

to obtain

$$\begin{aligned}
\mathcal{F}_{(2)}(\mu) &= \sum_{j,k} [\chi(\tau)]_{j,k} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_{j,k,h} \underbrace{[U^\dagger]_{j,h} [\gamma(\tau)]_h [U]_{h,k}}_{[\chi(\tau)]_{j,k}} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_{j,k,h} [\gamma(\tau)]_h \overline{[U]_{h,j}} [U]_{h,k} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_h [\gamma(\tau)]_h [2L_h \mu L_h^\dagger - \{L_h^\dagger L_h, \mu\}] \\
&= \sum_h [\gamma(\tau)]_h \mathcal{D}_{\sqrt{2}L_h}(\mu).
\end{aligned}$$

Because $\chi(\tau)$ is positive-semidefinite and thus $[\gamma(\tau)]_h$ are non-negative, then $\mathcal{F}_{(2)}$ is of Lindblad type. This implies

that the approximated time-dependent generator approximated at second order

$$\mathcal{F}_{t,2}(\mu) = -i[\tilde{H}, \mu] + \sum_h t[\gamma(\tau)]_h \mathcal{D}_{\sqrt{2}L_h}(\mu)$$

is also of Lindblad type at all times $t \geq 0$ as proved in Theorem 4.

Note that in deriving the reduced generator $\mathcal{F}_{t,2}$ there was no need to move the system's description into rotating frame. This provides another benefit of the proposed approach.

V. APPLICATIONS OF INTEREST

In this section we consider a few examples of interest: I) The first example is a classical linear model whose dynamics we want to project onto a subspace and showcases that the results derived in Sec. III do not only hold for quantum systems.

II) The second example considers a qubit coupled to a bosonic bath. As the exact solution of such a model is well known, this example allows us to assess the goodness of the approximation proposed in this work.

III) Third, a qubit coupled to a finite bath is studied. Given the fact that the bath is finite-dimensional this is typically considered an “highly non-Markovian model”. This example allows us to showcase the numerical procedure at higher orders for both a case where the original dynamics is purely Hamiltonian and a case where a dissipative term is included.

IV) Lastly an Ising spin chain is studied. This example allows us to showcase that the proposed procedure works in cases that are more general than the bipartite case and allows us to show that a non-Markovian reduced model can be obtained also for the population of the state (and not only for the bipartite case).

The code necessary to reproduce all the numerical experiment presented next is available at [31].

A. Linear testbed

To showcase the capability of the method we here consider a simple numerical example. Specifically, we consider a random autonomous and stable system of size

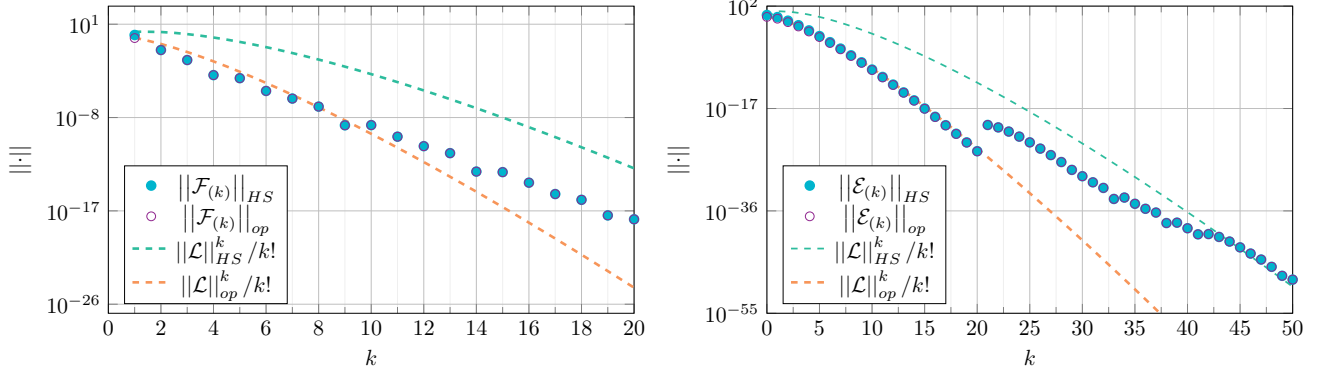


Figure 2: Operator and Hilbert-Schmidt norm of the first 20 $\mathcal{F}_{(k)}$ terms (left) and of the first 50 $\mathcal{E}_{(k)}$ terms when with $N = 20$ (right).

$n = 20$. Let then $\mathcal{A}$ be a random uniform matrix in $\mathbb{C}^{n \times n}$, i.e. $[\mathcal{A}]_{i,j} \sim \mathcal{U}([0,1])$ and let $\mathcal{B} = \mathcal{A} - \lambda_{\text{Max}} \mathbb{1}_{n \times n}$ where λ_{Max} is the absolute value of largest eigenvalue of $\mathcal{A}$. The dynamic's generator $\mathcal{L} \in \mathbb{C}^{n \times n}$ is then obtained as $\mathcal{L} = \frac{1}{2} \frac{\mathcal{B}}{\|\mathcal{B}\|_{op}}$ where $\|\cdot\|_{op}$ denotes the operator norm. This ensures that $\mathcal{L}$ is simply stable and has operator norm $\|\mathcal{L}\|_{op} = \frac{1}{2}$.

The projector $\mathcal{P}$ is then constructed through $\mathcal{R} = [\mathbb{1}_{m \times m} | 0_{m \times n-m}]$ and $\mathcal{J} = \mathcal{R}^\dagger$ to obtain

$$\mathcal{P} = \begin{bmatrix} \mathbb{1}_{m \times m} & 0_{m \times n-m} \\ 0_{m \times n-m} & 0_{n-m \times n-m} \end{bmatrix}.$$

By using Eq. (14) we can then compute the terms $\mathcal{F}_{(k)}$ iteratively for $k = 1, \dots, N$. Fig. 2 (left) shows both the Hilbert-Schmidt and operator norm of the terms $\mathcal{F}_{(k)}$ for $k = 1, 20$. Interestingly, the Hilbert-Schmidt norm, $\|\mathcal{F}_{(k)}\|_{HS}$, is bounded by $\frac{\|\mathcal{L}\|_{HS}^k}{k!}$, indicating that the series $\mathcal{F}_{t,N}$ should be converging for $N \rightarrow \infty$. Curiously, the same does not hold for the operator norm. Fig. 2 (right) instead shows the norm of the terms $\mathcal{E}_{(k)}$ where we fixed $N = 20$ and $\mathcal{F}_{(k)} = 0 \forall k > 20$. Interestingly for $k \leq N$ $\|\mathcal{E}_{(k)}\|$ is bounded by $\frac{\|\mathcal{L}\|_{op}^k}{k!}$ while for $k > N$ this is no longer the case.

Fig. 3 shows the error norm $\|\eta_t - \mu_t\|_2$ versus time. In this simulation η_t was simulated by reducing the original model, i.e. $\eta_t = \mathcal{R}e^{\mathcal{L}t}\rho_0$ while μ_t is obtained by simulating the time-dependent linear differential equation $\dot{\mu}_t = \mathcal{F}_{t,N}\mu_t$, for different values of N , by using an approximation of the equation given in Theorem 1, i.e. $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} \approx \sum_{k=0}^{100} \frac{t^k}{k!} \mathcal{L}^k \mathcal{J} \mu_0$. One can note that increasing the degree of approximation N decreases the error committed by the approximated model.

A natural alternative numerical approximation that comes to mind is given by truncating the exponential expansion at a given order N , i.e. $\eta_t \approx \sum_{k=0}^N \frac{t^k}{k!} \mathcal{R} \mathcal{L}^k \mathcal{J} \mu_0$. Note that such an approximation has numerical complexity almost identical to that of the approximation proposed in this work as in both cases one need to compute $\mathcal{L}^N$. Furthermore, from a theoretical standpoint, both

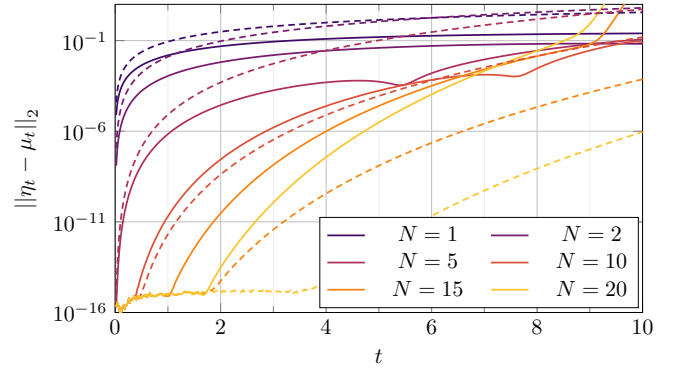


Figure 3: Error norm between the exact reduced evolution η_t and the approximated reduced evolution μ_t for different values of the approximation order N . The dashed curves represent an approximation obtained by truncating the Taylor expansion of the exponential for different values of the truncation order N .

cases commit an error of the order of $O(t^{N+1})$. To test this fact numerically, in Fig. 3 we included the actual error committed by these two approximations at the same orders of approximation: the continuous line denotes the error committed by the procedure proposed here while the dashed line is the error committed by the truncation of the exponential. One can observe that for lower orders ($N = 1, 2, 5$) the procedure proposed here yields better results, while at higher orders this is not the case.

B. Dephasing spin boson model

Inspired by [6, 32–34], in this section we consider a dephasing spin-boson model to further showcase the advances and drawbacks of the proposed method. Note that: i) in this section we make an exception to the assumption of final dimensional systems considered throughout the paper; ii) the model I am going to consider is exactly solvable hence the intent of this subsection

is to showcase that the proposed approximation method provides results comparable to other approximation techniques, not to compare it against the exact solution of the model. Nevertheless, this example helps us to understand the degrees of approximation that the proposed method offers and how the process unfolds.

The model is composed of a spin $\frac{1}{2}$ system $\mathcal{H}_S \simeq \mathbb{C}^2$ coupled to a bath $\mathcal{H}_B$ composed of N_B bosonic modes, $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. The model's dynamics is driven by the total Hamiltonian $H = H_S \otimes \mathbb{1}_B + \mathbb{1}_S \otimes H_B + H_{\text{int}}$ where

$$H_S = \frac{g}{2} \sigma_z \quad (21)$$

$$H_B = \sum_{k=1}^{N_B} \omega_k \left(n_k + \frac{1}{2} \right) \quad (22)$$

$$H_{\text{int}} = \sigma_z \otimes \left(\sum_{k=1}^{N_B} \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) \quad (23)$$

and where $n_k = b_k^\dagger b_k$ and b_k are the bosonic number and annihilation operator for mode k respectively and satisfy $[b_k, b_l^\dagger] = \delta_{k,l} \mathbb{1}$, $\omega_k \in \mathbb{R}$ is the mode frequency and $\lambda_k \in \mathbb{C}$ is the strength of the coupling between the mode and the spin.

We assume the initial state to be factorized $\rho_0 = \rho_{S,0} \otimes \rho_B$ with $\rho_B = e^{-\beta H_B} / \text{tr}[e^{-\beta H_B}]$ and some state $\rho_{S,0} \in \mathfrak{D}(\mathcal{H}_S)$ for the spin. We further assume the projector $\mathcal{P}$ to be $\mathcal{P}(\rho) = \text{tr}_B(\rho) \otimes \rho_B$ which can be factorized into $\mathcal{R}(\rho) = \text{tr}_B(\rho)$ and $\mathcal{J}(\sigma) = \sigma \otimes \rho_B$. One can then verify that $\mathcal{P}(\rho_0) = \rho_0$ and thus Assumption 1 is satisfied.

This model has a known exact analytical solutions given by [34, eq. (336) and (374)]:

$$\rho'_{S,t} = \begin{bmatrix} [\rho_{S,0}]_{0,0} & e^{-2\gamma(t)t} [\rho_{S,0}]_{0,1} \\ e^{-2\gamma(t)t} [\rho_{S,0}]_{1,0} & [\rho_{S,0}]_{1,1} \end{bmatrix}$$

where $\rho'_{S,t}$ is the spin's state in the rotating frame and $\gamma(t) = \sum_k \alpha_k \frac{1 - \cos(\omega_k t)}{t}$ with $\alpha_k \equiv 2 \frac{|\lambda_k|^2}{\omega_k^2} \coth\left(\frac{\beta \omega_k}{2}\right) \in \mathbb{R}$. By taking the derivative of an off diagonal element we obtain:

$$\frac{d}{dt} [\rho'_{S,t}]_{0,1} = -2 \underbrace{\left[\gamma(t) + \frac{d\gamma(t)}{dt} t \right]}_{\equiv \xi(t)} [\rho'_{S,t}]_{0,1}$$

where $\xi(t) = \sum_k \alpha_k \omega_k \sin(\omega_k t)$. This dynamics corresponds to the differential equation $\frac{d}{dt} \rho'_{S,t} = \xi(t) \mathcal{D}_{\sigma_z}[\rho'_{S,t}]$ for the state in rotating frame or, equivalently, to

$$\dot{\rho}_{S,t} = -i[H_S, \rho_{S,t}] + \xi(t) \mathcal{D}_{\sigma_z}(\rho_{S,t}) \quad (24)$$

for the state in the lab frame.

By following the procedure detailed in Sec. IV C we can then proceed to compute the reduced time-dependent generator at second order $\mathcal{F}_{t,2}$. Denoting

$\langle X \rangle_B \equiv \text{tr}[\rho_B X]$, we recall that [34, eq. (331)]:

$$\begin{aligned} \langle b_k^\dagger \rangle_B &= \langle b_k \rangle_B = \langle b_k^\dagger b_l^\dagger \rangle_B = \langle b_k b_l \rangle_B = 0, \\ \langle b_j^\dagger b_k \rangle_B &= \frac{\delta_{j,k}}{e^{\beta \omega_k} - 1}. \end{aligned}$$

We first compute the reduced Hamiltonian $\check{H} = \mathcal{J}^\dagger(H)$ where $\mathcal{J}^\dagger(X) = \text{tr}_B[X(\mathbb{1}_S \otimes \rho_B)]$. By noting that $\text{tr}_B[H_S \otimes \mathbb{1}_B(\mathbb{1}_S \otimes \rho_B)] = H_S \text{tr}[\rho_B] = H_S$, $\text{tr}_B[\mathbb{1}_S \otimes H_B(\mathbb{1}_S \otimes \rho_B)] = \mathbb{1}_S \langle H_B \rangle_B$ and

$$\begin{aligned} \text{tr}_B \left[\sigma_z \otimes \left(\sum_{k=1}^{N_B} \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) (\mathbb{1}_S \otimes \rho_B) \right] \\ = \sigma_z \left(\sum_{k=1}^{N_B} \lambda_k \langle b_k \rangle_B + \bar{\lambda}_k \langle b_k^\dagger \rangle_B \right) = 0 \end{aligned}$$

we have $\check{H} = H_S$ as the term $\mathbb{1} \langle H_B \rangle_B$ only contributes to a phase and can thus be removed leading to the first order term $\mathcal{F}_{(1)}(\mu) = -i[\check{H}, \mu]$. The second order term is instead found to be $\mathcal{F}_{(2)}(\mu) = \varphi \mathcal{D}_{\sigma_z}(\mu)$ with $\varphi \equiv \sum_{k=1}^{N_B} \frac{2|\lambda_k|^2}{e^{\beta \omega_k} - 1}$. All the necessary calculations are detailed in Appendix B for the sake of clarity.

Combining both terms we obtain an approximate reduced model described by the time-dependent Lindblad equation

$$\dot{\mu}_t = -i[H_S, \mu_t] + \varphi t \mathcal{D}_{\sigma_z}(\mu_t). \quad (25)$$

By comparing this equation with the exact solution of Eq. (24) we can observe that both the Hamiltonian and noise operator are correctly recovered whereas the time-dependent dissipation strength $\xi(t)$ is approximated with φt . Interestingly, the value φ does not coincide with the first order Taylor expansion of $\xi(t)$ which is $\sum_k \alpha_k \omega_k$.

To better assess the quality of the reduced second order approximation we next compare the evolution of $\rho_{S,t}$ and that of μ_t with the coarse-grained approximation discussed in [34, eq. (332)], i.e. $\tilde{\rho}_{S,t} \in \mathfrak{D}(\mathcal{H}_S)$ and

$$\dot{\tilde{\rho}}_{S,t} = -i[H_S, \tilde{\rho}_{S,t}] + \gamma(\tau) \mathcal{D}_{\sigma_z}(\tilde{\rho}_{S,t}) \quad (26)$$

where the constant $\tau \in \mathbb{R}$ is specified later.

The bath coupling coefficients are going to be defined as $\lambda_k = \sqrt{J(\omega_k)}$ [35] where $J(\omega)$ is the bath spectral density and is chosen to be

$$J(\omega) = \Lambda \omega_c^{1-s} \omega^s e^{-\frac{\omega}{\omega_c}} \quad (27)$$

where $\Lambda \in \mathbb{R}_+$ is an overall constant, $\omega_c \in \mathbb{R}_+$ is the bath cutoff frequency and $s \in \mathbb{R}_+$ is the Ohmicity parameter which determines three cases of interest, i.e. Ohmic ($s = 1$), sub-Ohmic ($s < 1$) and super-Ohmic ($s > 1$) [36].

In Fig. 4 the results of the simulations are shown in the three cases, i.e. $s = 0.5, 1, 1.5$. The other parameters used in the simulations are $g = 1.8$, $\omega_c = 0.2$, $\Lambda = 0.5$, $\beta = 10$, $\tau = 32/\omega_c$, and where ω_k are $N_B = 100$, taken

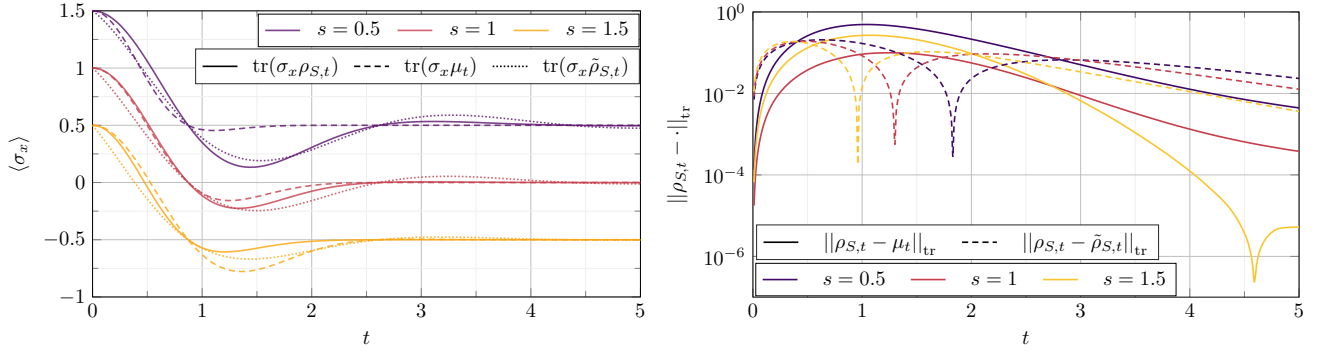


Figure 4: Simulation of the dephasing spin-boson model for $s = 0.5, 1, 1.5$ where we compare the exact solution $\rho_{S,t}$ given in Eq.(24), the second order approximation μ_t derived in this work, given in Eq. (25) and the coarse-grained dynamics $\tilde{\rho}_{S,t}$ given in Eq. (26). Left: Evolution of the expectation value $\langle \sigma_x \rangle$ for the exact solution (continuous curves), second order approximation (dashed curve) and coarse-grained dynamics (dotted curve). The expectations values of the cases $s = 0.5, 1.5$ have been shifted by ± 0.5 respectively for graphical purposes. Right: evolution of the trace-norm error with respect to the exact solution committed by the second order approximation (continuous curve) and coarse-grained dynamics (dashed curve).

equally spaced between 0 and 1. The initial state instead has been fixed to $\rho_{S,0} = \mu_0 = \tilde{\rho}_{S,0} = |+\rangle\langle +|$. Importantly, one can observe in Fig. 4 Right that, in each of the three cases, the error committed by the second order approximation, Eq. (25) is below that of the coarse-grained dynamics, Eq. (26) for both the initial and final phases while it is above for a central phase.

This indicates that the second order approximation performs better than the coarse-grained Markovian approximation in this case.

C. Dissipative central spin model

In this section we consider a dissipative central spin model to showcase the capabilities of the proposed technique. Similar models have been studied in the literature, see e.g. [37–40]. The model is composed by a system, i.e. $\mathcal{H}_S \simeq \mathbb{C}^2$ also denoted as spin 1 and a bath composed of N_B other spins, i.e. $\mathcal{H}_B \simeq \mathbb{C}^{2^{N_B}}$ counted from 2 to $N_B + 1$.

The system Hamiltonian is described as $H = H_S + H_B + H_{\text{int}}$ where:

$$\begin{aligned} H_S &= \delta(\sigma_x^{(1)} + \sigma_z^{(1)}) \\ H_B &= \frac{\gamma}{4} \left(2J_x^2 - \frac{N_B}{2} \mathbb{1} \right) \\ H_{\text{int}} &= \frac{\lambda}{2} \left(A_x \sigma_x^{(1)} J_x + A_y \sigma_y^{(1)} J_y + A_z \sigma_z^{(1)} J_z \right) \end{aligned}$$

where $\sigma_u^{(j)}$ denotes the $u \in \{x, y, z\}$ Pauli matrix acting on the j -th spin and $J_u = \frac{1}{2} \sum_{k=2}^{N_B+1} \sigma_u^{(k)}$ denotes the total bath angular momentum operators. The bath is also in contact with a Markovian environment that dissipates information. This interaction is described by the noise

operators $L_k = \Lambda \sigma_-^{(k)}$ for all $k = 2, \dots, N_B + 1$ where $\sigma_{\pm}^{(k)} = \frac{1}{2}(\sigma_x^{(k)} \pm \sigma_y^{(k)})$.

For this example we consider as an initial condition the factorized state $\rho_0 = |+\rangle\langle +| \otimes \rho_B$ where

$$\rho_B = \frac{e^{-\beta H_B}}{\text{tr}[e^{-\beta H_B}]}$$

is the thermal state at inverse temperature β . This allows us to define the reduction and injection maps as $\mathcal{R}(\cdot) = \text{tr}_B(\cdot)$ and $\mathcal{J}(\cdot) = \cdot \otimes \rho_B$. Note that with this choice we have that $\mathcal{P}(\rho_0) = \rho_0$ hence Assumption 1 is satisfied. The reduced state then corresponds to the state of the central spin, i.e.

$$\eta_t = \rho_S(t) \equiv \text{tr}_B(\rho_t).$$

In Fig.5 the results of the simulation for this dissipative central spin model are shown. The simulation has been run in two settings: a purely Hamiltonian setting ($\Lambda = 0$ on the left) and a dissipative setting ($\Lambda = 0.8$ on the right). The rest of the parameters used were $N_B = 3$, $\delta = 0.3$, $\lambda = 0.1$, $\gamma = 1$, $A_x = 1.2$, $A_y = 1.5$, $A_z = 1.3$ and $\beta = 50$. In the numerical simulations we can observe a few interesting aspects: first, as expected, the error between the exact solution and the approximate solution $\|\eta_t - \mu_t\|_t$ decreases when increasing the approximation degree N ; second, the trajectory for approximation orders $N = 1, 2$ and even 3 do not exit the Bloch sphere (note that we have proven the complete positivity for orders $N = 1, 2$ but not for $N = 3$); the simulations for $N > 3$ are displaying Runge's phenomena at larger times, which is quite typical in these situations; after some time, the evolved state η_t for an approximation degree $N \geq 4$ exits the Bloch sphere, indicating that the dynamics is not CPTP at all times for approximation degrees higher than $N \geq 4$.

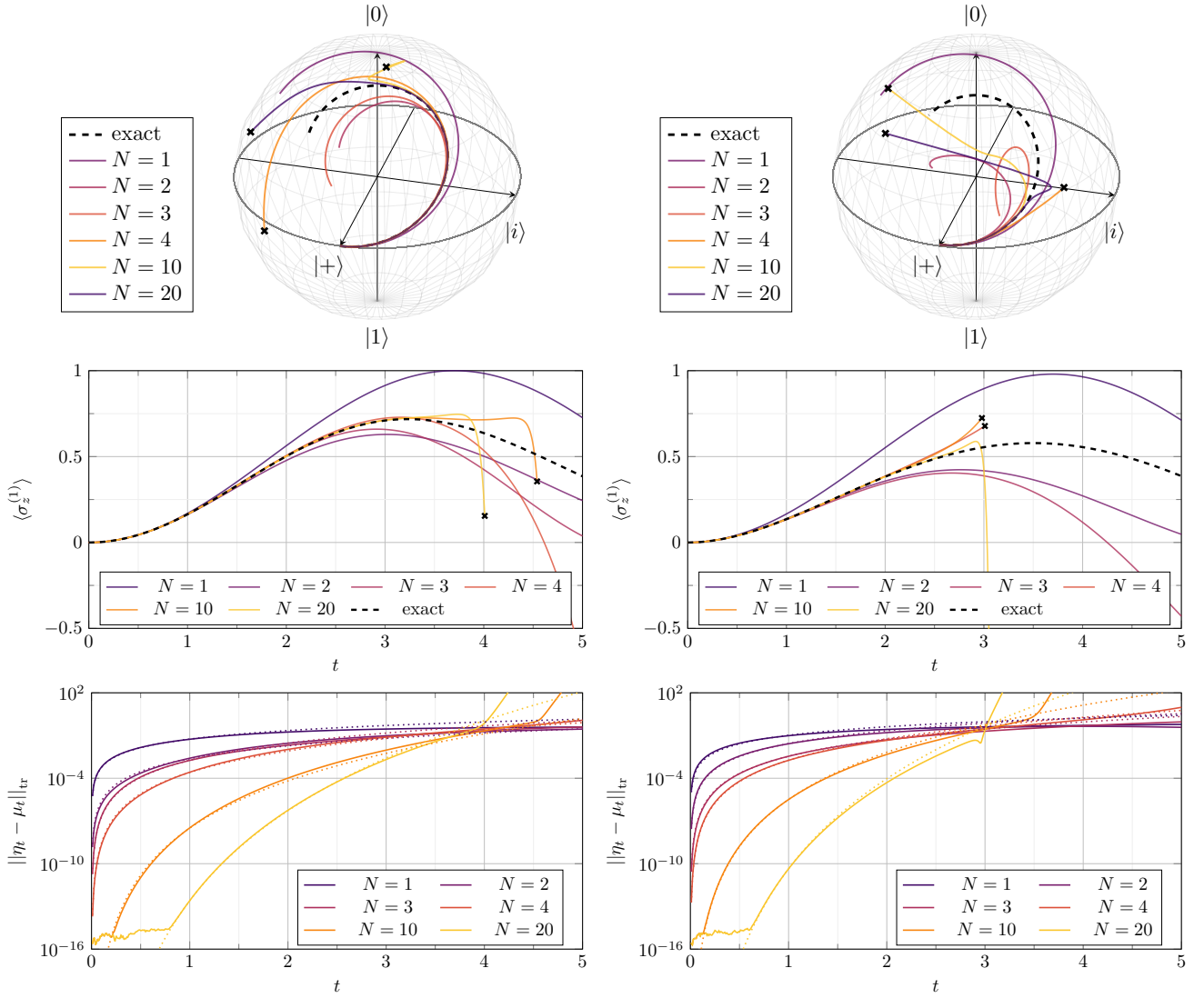


Figure 5: Simulations of the dissipative central spin model in the purely Hamiltonian setting ($\Lambda = 0$) on the left and in the dissipative setting ($\Lambda = 0.8$) on the right. The exact evolution and various degrees of approximations ($N = 1, 2, 3, 4, 10, 20$) are shown in each figure. The first row shows the trajectory of the central spin's state ρ_S in the Bloch sphere; the second row shows the magnetization of the central spin in the σ_z direction versus time; the third row shows the trace norm error between the exact trajectory η_t and the reduced one μ_t , i.e. $\|\eta_t - \mu_t\|_{\text{tr}}$ versus time for different values of the approximation N and with dotted curve representing a fitted αt^{N+1} curve. The black crosses denote points where the evolved state μ_t exits the set of density operators.

D. Ising spin chain

In order to show that the proposed method works also in cases that are not bi-partite we here consider a spin-chain model. More specifically we consider a model composed of N_B spins, i.e. $\mathcal{H} \simeq \mathbb{C}^n$ where $n = 2^{N_B}$ whose dynamics is driven by the Ising Hamiltonian

$$H = \sum_{j=1}^{N_B} h \sigma_z^{(j)} + A \sum_{j=1}^{N_B-1} \sigma_x^{(j)} \sigma_x^{(j+1)}. \quad (28)$$

In this example we consider $\mathcal{P}$ to be the CPTP projector onto the diagonal $\mathcal{P}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ which can be factorized into $\mathcal{R}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ and $\mathcal{J}(\eta) = \sum_{j=0}^{n-1} |j\rangle\langle j| \langle j| \eta \rangle$. Note that, in such a case, $\eta = \mathcal{R}(\rho) \in \mathbb{R}^n$ is a probability distribution, i.e. $\eta_i \geq 0$ and $\mathbf{1}^T \eta = 1$ where $\mathbf{1} \in \mathbb{R}^n$ is the vector of all ones. This implies that $\mathcal{F}_t \in \mathbb{R}^{n \times n}$.

As an initial condition we consider $\rho_0 = |0\rangle\langle 0|$. Trivially, $\mathcal{P}(\rho_0) = \rho_0$ hence Assumption 1 is satisfied.

Numerically, we verified that $\mathcal{F}_{(1)} = \mathcal{F}_{(3)} = 0$ while $\mathcal{F}_{(2)}$ is graphically represented in Fig. 7 where one can

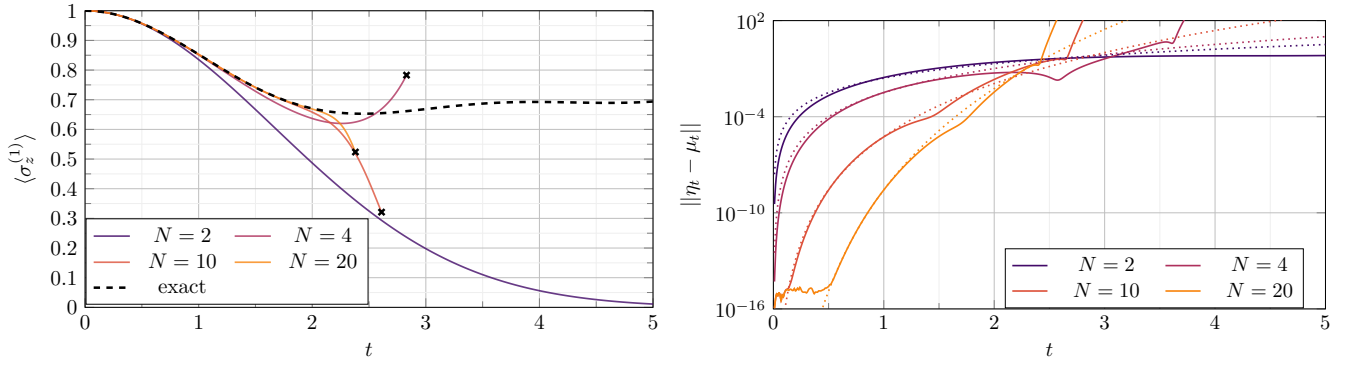


Figure 6: Simulation of the Ising spin chain. Magnetization of the first spin of the chain $\langle \sigma_z^{(1)} \rangle$ versus time on the left and the error norm $\|\eta_t - \mu_t\|_2$ versus time on the right. Various approximation's degrees $N = 2, 4, 10, 20$ are shown in each figure. The black crosses denote the points where the evolved state μ_t exits the n -dimensional simplex, i.e. μ_t is no longer a probability distribution while the dotted curves on the right again represent fitted curves of αt^{N+1} .

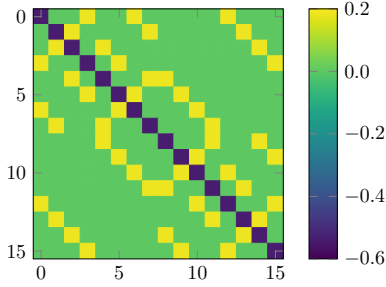


Figure 7: Graphical representation of the term $\mathcal{F}_{(2)}$ for the time-dependent generator $\mathcal{F}_t$ of the Ising spin chain model.

verify that $\mathcal{F}_{(2)}$ is Metzler (i.e. A matrix A is Metzler if $A_{i,j} \geq 0$ for all $i \neq j$) and is such that $\mathbf{1}^T \mathcal{F}_{(2)} = 0$. Those two conditions are the classical equivalent to the properties characterizing a Lindblad generator and in fact imply that $e^{\mathcal{F}_{(2)}t}$ is a stochastic matrix for all $t \geq 0$. This directly implies that $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,2} ds}$ is CPTP, or stochastic in this case, for all $t \geq 0$.

Fig. 6 shows the results of the simulations of the Ising spin chain model with $N_B = 4$, $A = 0.3$, $h = 0.36$ where we can observe similar results to the other simulations included in this work: increasing the approximation degree decreases the error $\|\eta_t - \mu_t\|_2$ at low times; for $N = 2$ the approximate evolved state μ_t remains a probability distribution while for $N \geq 4$, after a certain time, the state is no longer a probability distribution.

VI. CONCLUSIONS AND OUTLOOK

Pros of the method: 1) The method is iterative and can thus be easily tested (implemented) for small system's sizes. 2) Does not require moving to a rotating

frame. 3) The method does not only work for Hamiltonian methods but also includes Lindbladian evolutions. 4) The approach is very general and does not only include the

Cons of the method: 1) The method only works for short-time approximations.

Outlook:

- convergence of the considered series (lifting part Assumption 2 to get from polynomial to analytic). Say it can be interesting mathematically but practically one would not want to do it. Also explain the difficulties we could encounter (proof of thm1).
- What happens if we remove the assumption $\mathcal{P}(\rho_0) = \rho_0$

VII. CODE AVAILABILITY

The code necessary to reproduce all the numerical experiments presented in this work is available at [31].

VIII. ACKNOWLEDGMENTS

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Appendix A: Technical results and proofs

1. Time-ordered expansion

Before proving the main result, Theorem 1, we shall develop some intuition about how we prove it.

Example 1. To provide some intuition about Theorem 1 let us consider a simple case: $\mathcal{F}_t = \mathcal{F}_{(1)} + t\mathcal{F}_{(2)} + t^2\mathcal{F}_{(3)}$. Let us first compute the first few terms of the Dyson expansion [41] of $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$, i.e. $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t}$. At zeroth order we have $\Phi_{(0),t} = \mathcal{I}$ while at orders 1, 2 and 3 we have:

$$\begin{aligned} \Phi_{(1),t} &= \int_0^t \mathcal{F}_s ds = \int_0^t (\mathcal{F}_{(1)} + s\mathcal{F}_{(2)} + s^2\mathcal{F}_{(3)}) ds = t\mathcal{F}_{(1)} + t^2 \frac{\mathcal{F}_{(2)}}{2} + t^3 \frac{\mathcal{F}_{(3)}}{3}, \\ \Phi_{(2),t} &= \int_0^t dt_1 \int_0^{t_1} dt_2 \mathcal{F}_{t_1} \mathcal{F}_{t_2} = \sum_{j=0}^1 \int_0^t dt_1 \mathcal{F}_{t_1} \mathcal{F}_{(j+1)} \int_0^{t_1} dt_2 t_2^j = \sum_{j=0}^1 \int_0^t dt_1 \mathcal{F}_{t_1} \mathcal{F}_{(j+1)} \frac{t_1^{j+1}}{j+1} \\ &= \sum_{k,j=0}^1 \mathcal{F}_{(k+1)} \mathcal{F}_{(j+1)} \int_0^t dt_1 t_1^k \frac{t_1^{j+1}}{j+1} = \sum_{k,j=0}^1 \mathcal{F}_{(k+1)} \mathcal{F}_{(j+1)} \frac{t^{k+j+2}}{(k+j+2)(j+1)} = \sum_{k,j=1}^2 \mathcal{F}_{(k)} \mathcal{F}_{(j)} \frac{t^{k+j}}{(j+k)j} \\ &= \underbrace{\mathcal{F}_{(1)}^2 \frac{t^2}{2}}_{j=k=1} + \underbrace{\mathcal{F}_{(2)} \mathcal{F}_{(1)} \frac{t^3}{3}}_{j=1,k=2} + \underbrace{\mathcal{F}_{(1)} \mathcal{F}_{(2)} \frac{t^3}{6}}_{j=2,k=1} + O(t^4), \\ \Phi_{(3),t} &= \int_0^t dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \mathcal{F}_{t_1} \mathcal{F}_{t_2} \mathcal{F}_{t_3} = \int_0^t dt_1 \mathcal{F}_{t_1} \underbrace{\int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \mathcal{F}_{t_2} \mathcal{F}_{t_3}}_{=\sum_{j,k=1}^2 \mathcal{F}_{(k)} \mathcal{F}_{(j)} \frac{t_1^{k+j}}{(j+k)j}} \\ &= \sum_{h,k,j=1}^2 \mathcal{F}_{(h)} \mathcal{F}_{(k)} \mathcal{F}_{(j)} \int_0^t dt_1 t_1^h \frac{t_1^{k+j}}{(j+k)j} = \sum_{h,k,j=1}^2 \mathcal{F}_{(h)} \mathcal{F}_{(k)} \mathcal{F}_{(j)} \frac{t^{h+k+j}}{(h+k+j)(j+k)j} = \underbrace{\mathcal{F}_{(1)}^3 \frac{t^3}{3!}}_{h=k=j=1} + O(t^4). \end{aligned}$$

Grouping together the terms of $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$ by the order of the exponent of t , we have

$$\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \underbrace{\mathcal{I}}_{\mathcal{E}_{(0)}} + t \underbrace{\mathcal{F}_{(1)}}_{\mathcal{E}_{(1)}} + t^2 \underbrace{\left(\frac{\mathcal{F}_{(2)}}{2} + \frac{\mathcal{F}_{(1)}^2}{2} \right)}_{\mathcal{E}_{(2)}} + t^3 \underbrace{\left(\frac{\mathcal{F}_{(3)}}{3} + \frac{\mathcal{F}_{(1)} \mathcal{F}_{(2)}}{6} + \frac{\mathcal{F}_{(2)} \mathcal{F}_{(1)}}{3} + \frac{\mathcal{F}_{(1)}^3}{3!} \right)}_{\mathcal{E}_{(3)}} + O(t^4)$$

which coincides with the first few terms obtained using the equation given in Theorem 1:

$$\begin{aligned}\mathcal{E}_{(0)} &= \mathcal{I}, \\ \mathcal{E}_{(1)} &= \mathcal{F}_{(1)}\mathcal{E}_{(0)} = \mathcal{F}_{(1)}, \\ \mathcal{E}_{(2)} &= \frac{1}{2}(\mathcal{F}_{(1)}\mathcal{E}_{(1)} + \mathcal{F}_{(2)}\mathcal{E}_{(0)}) = \frac{1}{2}(\mathcal{F}_{(1)}^2 + \mathcal{F}_{(2)}), \\ \mathcal{E}_{(2)} &= \frac{1}{3}(\mathcal{F}_{(1)}\mathcal{E}_{(2)} + \mathcal{F}_{(2)}\mathcal{E}_{(1)} + \mathcal{F}_{(3)}\mathcal{E}_{(0)}) = \frac{1}{3}\left(\frac{1}{2}\mathcal{F}_{(1)}^3 + \frac{1}{2}\mathcal{F}_{(1)}\mathcal{F}_{(2)} + \mathcal{F}_{(2)}\mathcal{F}_{(1)} + \mathcal{F}_{(3)}\right).\end{aligned}$$

Proof (Theorem 1). We shall start by proving that $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t}$ where $\Phi_{(0),t} = \mathcal{I}$ and, for $n > 0$,

$$\Phi_{(n),t} = \sum_{k_1, \dots, k_n=1}^{N+1} t^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

First of all, using the Dyson series expansion for the time-ordered exponential $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$, see e.g. [41–43], we have that:

$$\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t} \quad \text{with} \quad \Phi_{(n),t} = \int_0^t dt_1 \int_0^{t_1} dt_2 \dots \int_0^{t_{n-1}} dt_n \mathcal{F}_{t_1} \mathcal{F}_{t_2} \dots \mathcal{F}_{t_n}.$$

Next, by noticing that $\Phi_{(n),t} = \int_0^t ds \mathcal{F}_s \Phi_{(n-1),s}$ we can prove the statement by induction. The case $n = 1$ is proven by simple calculations:

$$\Phi_{(1),t} = \int_0^t ds \mathcal{F}_s = \int_0^t ds \sum_{k_1=0}^N s^{k_1} \mathcal{F}_{(k_1+1)} = \sum_{k_1=0}^N \mathcal{F}_{(k_1+1)} \int_0^t ds s^{k_1} = \sum_{k_1=0}^N \mathcal{F}_{(k_1+1)} \frac{t^{k_1+1}}{k_1+1} = \sum_{k_1=1}^{N+1} \mathcal{F}_{(k_1)} \frac{t^{k_1}}{k_1}.$$

Note that $\int_0^t ds \sum_{k=0}^N = \sum_{k=0}^N \int_0^t ds$ under the assumption that N is finite, i.e. $N < \infty$. We next assume that the statement holds for n , i.e.

$$\Phi_{(n),t} = \sum_{k_1, \dots, k_n=1}^{N+1} t^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

Then

$$\begin{aligned}
\Phi_{(n+1),t} &= \int_0^t ds \mathcal{F}_s \Phi_{(n),s} = \int_0^t ds \mathcal{F}_s \sum_{k_1, \dots, k_n=1}^{N+1} s^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_n)} \dots \mathcal{F}_{(k_1)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\
&= \int_0^t ds \sum_{k=0}^N s^k \mathcal{F}_{(k+1)} \sum_{k_1, \dots, k_n=1}^{N+1} s^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\
&= \sum_{k=0}^N \sum_{k_1, \dots, k_n=1}^{N+1} \mathcal{F}_{(k+1)} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \int_0^t s^{k+\sum_{j=1}^n k_j} ds \\
&= \sum_{k=0}^N \sum_{k_1, \dots, k_n=1}^{N+1} \frac{t^{k+1+\sum_{j=1}^n k_j}}{k+1+\sum_{j=1}^n k_j} \mathcal{F}_{(k+1)} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\
&= \sum_{k=1}^{N+1} \sum_{k_1, \dots, k_n=1}^{N+1} \frac{t^{k+\sum_{j=1}^n k_j}}{k+\sum_{j=1}^n k_j} \mathcal{F}_{(k)} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\
&= \sum_{k_1=1}^{N+1} \sum_{k_2, \dots, k_{n+1}=1}^{N+1} \frac{t^{k_1+\sum_{j=2}^{n+1} k_j}}{k_1+\sum_{j=2}^{n+1} k_j} \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \dots \mathcal{F}_{(k_{n+1})}}{\prod_{j=1}^n \sum_{h=n-j+2}^{n+1} k_h} \\
&= \sum_{k_1, \dots, k_{n+1}=1}^{N+1} \frac{t^{\sum_{j=1}^{n+1} k_j}}{\sum_{j=1}^{n+1} k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_{n+1})}}{\prod_{j=1}^n \sum_{h=n-j+2}^{n+1} k_h} \\
&= \sum_{k_1, \dots, k_{n+1}=1}^{N+1} t^{\sum_{j=1}^{n+1} k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_{n+1})}}{\prod_{j=1}^{n+1} \sum_{h=n-j+2}^{n+1} k_h}.
\end{aligned}$$

thus proving the statement.

We next prove that $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{f=0}^{+\infty} t^k \mathcal{E}_{(k)}$ where $\mathcal{E}_{(0)} = \mathcal{I}$ and, for $k > 0$,

$$\mathcal{E}_{(k)} = \sum_{n=1}^k \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^k \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

To prove this claim we can start by noticing that each term $\Phi_{(n),t}$ contains powers t^k of order $k \geq n$. Hence, to construct $\mathcal{E}_{(k)}$ of order k we need to collect and sum all the coefficients that multiply t^k among the terms $\Phi_{(n),t}$ for $1 \leq n \leq k$. Given $\Phi_{(n),t}$ with $n \leq k$ we have that the coefficient multiplying t^k must satisfy the constraint $\sum_{j=1}^n k_j = k$. Thus, the coefficient of $\Phi_{(n),t}$ that multiply t^k are

$$\sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^{N+1} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} = \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^k \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}$$

where we used the fact that, if any $k_j > k$ then $\sum_{j=1}^n k_j > k$. Noticing that $\prod_{j=1}^n \sum_{h=n-j+1}^n k_h = k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)$ and summing over $1 \leq n \leq k$ we obtain, for $k > 0$, the second claim, i.e.

$$\mathcal{E}_{(k)} = \sum_{n=1}^k \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^k \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)}.$$

We are now finally ready to prove the main claim of the theorem, that is: $\mathcal{E}_{(k)} = \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)}$. Recalling that

$\mathcal{E}_{(0)} = \mathcal{I}$ we can write:

$$\begin{aligned} \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} &= \sum_{s=1}^{k-1} \frac{\mathcal{F}_{(s)}}{k} \mathcal{E}_{(k-s)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{s=1}^{k-1} \sum_{n=1}^{k-s} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{(k-s) (\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{s=1}^{k-1} \sum_{n=1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{(k-s) (\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \end{aligned}$$

where we used the fact that $\sum_{n=k-s+1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{(k-s) (\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} = 0$ since for $n > k-s$ we have that $\sum_{j=1}^n k_j > k-s$ for any set of k_j . This implies that

$$\begin{aligned} \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} &= \sum_{n=1}^k \sum_{s=1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{(k-s) (\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=1}^{k-1} \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_{n+1}=1 \\ \text{s.t. } \sum_{j=2}^{n+1} k_j = k-k_1}}^k \frac{\mathcal{F}_{(k_1)}}{k} \frac{\mathcal{F}_{(k_2)} \dots \mathcal{F}_{(k_{n+1})}}{\underbrace{(k-k_1) (\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_{h+1})}_{\sum_{h=2}^{n+1} k_h}} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=1}^{k-1} \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_{n+1}=1 \\ \text{s.t. } \sum_{j=1}^{n+1} k_j = k}}^k \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \dots \mathcal{F}_{(k_{n+1})}}{k (\prod_{j=1}^n \sum_{h=n-j+2}^{n+1} k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=2}^k \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^k \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \dots \mathcal{F}_{(k_n)}}{k (\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=1}^k \sum_{k_1=1}^k \sum_{\substack{k_2, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^k \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \dots \mathcal{F}_{(k_n)}}{k (\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} = \mathcal{E}_{(k)}, \end{aligned}$$

thus concluding the proof. $\square$

Appendix B: Calculations for the spin boson model

We can write

$$H = \underbrace{\mathbb{1}_S}_{\equiv S_0} \otimes \underbrace{H_B}_{\equiv E_0} + \underbrace{\sigma_z}_{S_1} \otimes \underbrace{\left(\frac{g}{2} \mathbb{1} + \sum_k \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right)}_{\equiv E_1}$$

so that the second order term becomes $\mathcal{F}_{(2)}(\mu) = \sum_{j,k=0}^1 \chi_{j,k}(\rho_B) \left(S_j \mu S_k - \frac{1}{2} \{S_j^\dagger S_k, \mu\} \right)$ where $\chi_{j,k}(\rho_B) = \langle E_k E_j \rangle_B - \langle E_j \rangle_B \langle E_k \rangle_B$. We can then start by noting that $\langle E_1 \rangle_B = g/2$ and

$$\langle H_B \rangle_B = \sum_{k=1}^{N_B} \omega_k \left(\langle b_k^\dagger b_k \rangle_B + \langle \mathbb{1} \rangle_B / 2 \right) = \sum_{k=1}^{N_B} \omega_k \left(\frac{1}{e^{\beta \omega_k} - 1} + \frac{1}{2} \right) = \sum_{k=1}^{N_B} \frac{\omega_k}{2} \frac{e^{\beta \omega_k} + 1}{e^{\beta \omega_k} - 1}$$

in order to compute

$$\begin{aligned}
\chi_{1,0}(\rho_B) &= \left\langle \left(\frac{g}{2} \mathbb{1} + \sum_j \lambda_j b_j + \bar{\lambda}_j b_j^\dagger \right) \sum_k \omega_k \left(n_k + \frac{1}{2} \right) \right\rangle_B - \frac{g}{2} \langle H_B \rangle_B \\
&= \sum_k \frac{g}{2} \omega_k (\langle n_k \rangle_B + \frac{1}{2}) + \sum_j \lambda_j \omega_k (\cancel{\langle b_j n_k \rangle_B} + \cancel{\langle b_j \rangle_B}) + \bar{\lambda}_j \omega_k (\cancel{\langle b_j^\dagger n_k \rangle_B} + \cancel{\langle b_j^\dagger \rangle_B}) - \frac{g}{2} \langle H_B \rangle_B \\
&= \sum_k \frac{g}{2} \omega_k \left(\frac{1}{e^{\beta \omega_k} - 1} + \frac{1}{2} \right) - \frac{g}{2} \frac{\omega_k e^{\beta \omega_k} + 1}{e^{\beta \omega_k} - 1} = 0
\end{aligned}$$

as $\langle b_j n_k \rangle_B = \langle b_j^\dagger n_k \rangle_B = 0$.

This fact directly implies that $\mathcal{F}_{(2)}(\mu) = \sum_{j=0}^1 \chi_{j,j}(\rho_B) \mathcal{D}_{S_j}(\mu)$ and, since $S_0 = \mathbb{1}_S$ we have that $\mathcal{D}_{S_0} = 0$ thus recovering $\mathcal{F}_{(2)}(\mu) = \chi_{1,1}(\rho_B) \mathcal{D}_{\sigma_z}(\mu)$. It then remains to compute the coefficient

$$\begin{aligned}
\chi_{1,1}(\rho_B) &= \left\langle \left(\frac{g}{2} \mathbb{1} + \sum_k \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) \left(\frac{g}{2} \mathbb{1} + \sum_k \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) \right\rangle_B - \frac{g^2}{4} \\
&= \cancel{\frac{g^2}{4} \langle \mathbb{1} \rangle_B} + \sum_k g \left(\lambda_k \cancel{\langle b_k \rangle_B} + \bar{\lambda}_k \cancel{\langle b_k^\dagger \rangle_B} \right) \\
&\quad + \sum_{j,k} \left(\lambda_j \lambda_k \cancel{\langle b_j b_k \rangle_B} + \lambda_j \bar{\lambda}_k \langle b_j b_k^\dagger \rangle_B + \bar{\lambda}_j \lambda_k \langle b_j^\dagger b_k \rangle_B + \bar{\lambda}_j \bar{\lambda}_k \cancel{\langle b_j^\dagger b_k^\dagger \rangle_B} \right) - \cancel{\frac{g^2}{4}} \\
&= \sum_{k=1}^{N_B} \frac{2|\lambda_k|^2}{e^{\beta \omega_k} - 1} \equiv \varphi.
\end{aligned}$$